



# Accelerate Test Execution with Adaptive Test

Reference: RA\_453

## Unlock Smarter Testing with DAS

Welcome to the Adaptive Test DAS Demo — a powerful illustration of how test programs can be made smarter, more flexible, and responsive to real-time conditions. Whether you're running it locally or envisioning a scaled deployment on remote servers, this demo is built to showcase adaptability.

## What This Demo Demonstrates

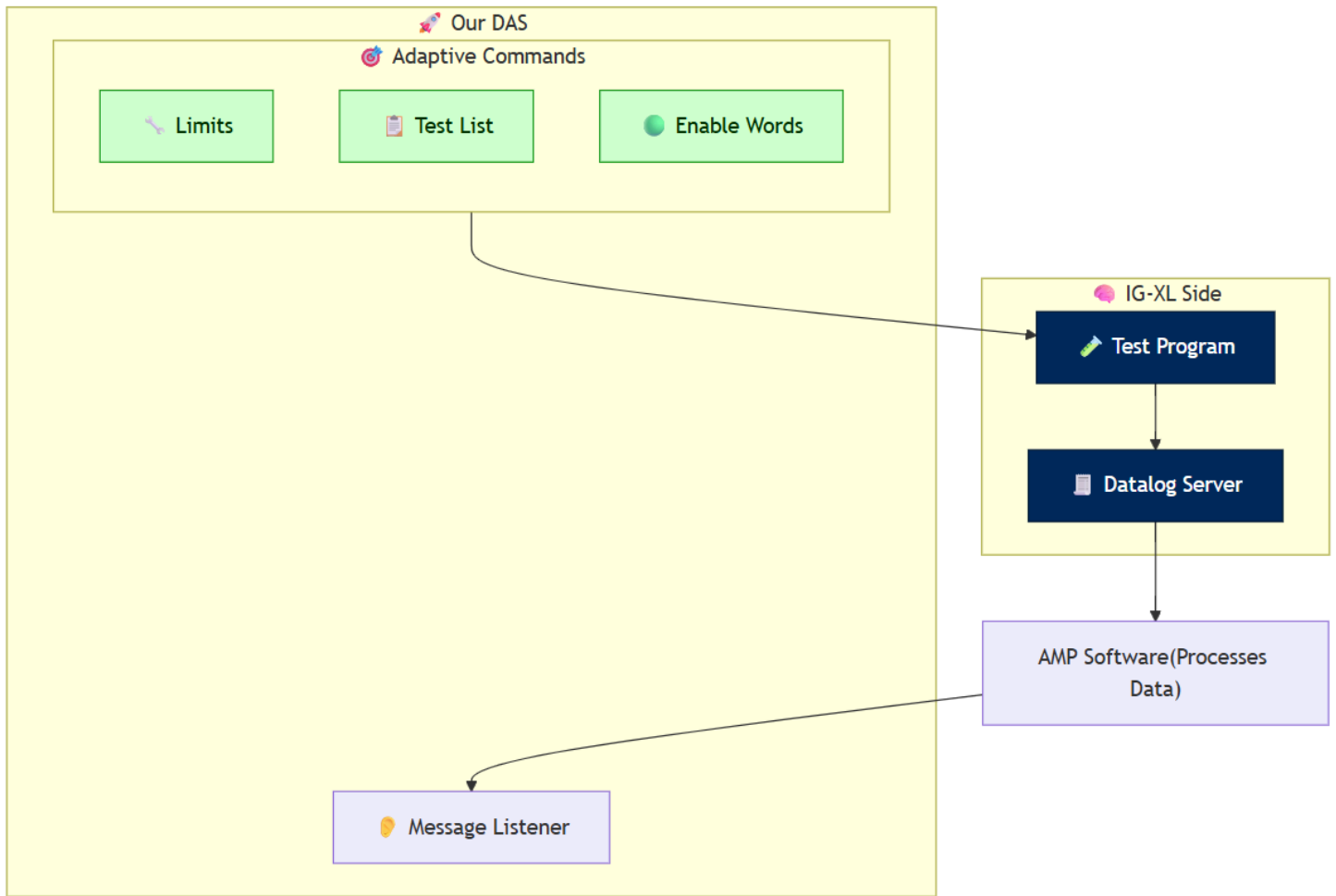
This demo highlights three key capabilities of our Data Analytics Solution (DAS):

1. **Enabling or disabling specific tests** — streamline test execution based on context or priorities
2. **Modifying test limits dynamically** — improve accuracy and adapt to production variability
3. **Controlling enable words** — activate or deactivate sub-flows conditionally

Additionally, the system offers robust observability:

- Monitor messages sent by the tester in real time
- Understand communication patterns driven by adaptive logic

## Software Overview



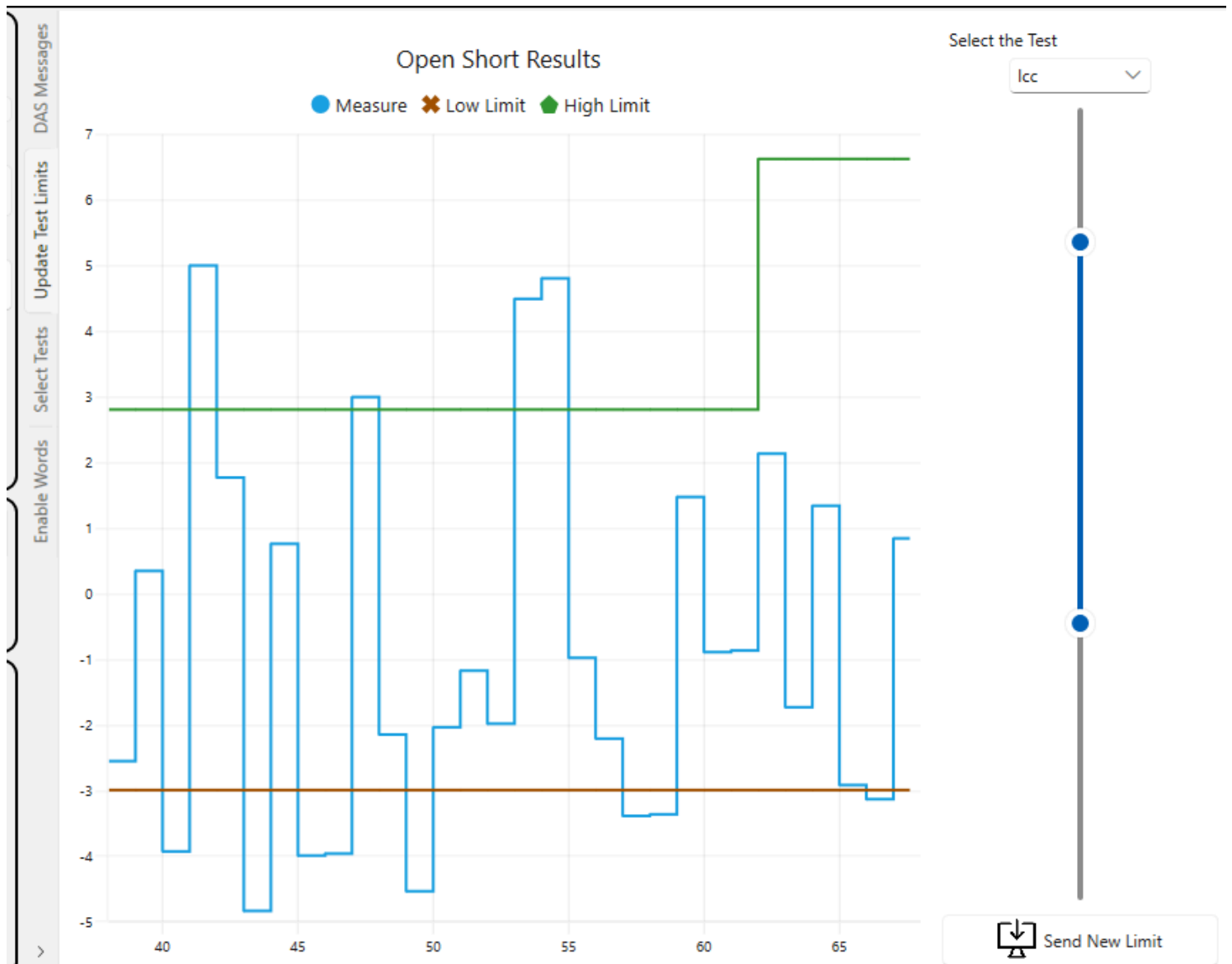
## Explore the Demo Software

The demo application includes four dedicated sections:

### DAS Section

View all messages generated by the tester. Clicking on a message reveals its detailed contents.





## Enable Word Section

The test program includes two enable words that control sub-flow activation. Enable or disable them directly from this interface.



| Component   | Version       |
|-------------|---------------|
| Framework   | .NET 8        |
| Environment | Windows 10    |
| Excel       | 2016 (64-bit) |
| IGXL        | 10.60.02      |

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## Extend the Demo to Your Environment

To implement a similar system, we recommend reviewing the following guides:

| Reference Architecture | Description                           |
|------------------------|---------------------------------------|
| RA_0468                | How to implement a DAS in C#          |
| RA_0478                | How to send an adaptive command in C# |

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## See It in Action

Explore the complete [Instructions to Run the Demo](#) or browse the image gallery to discover key functionalities.

Looking to integrate this approach into your test strategy? [Contact our team](#) to explore how DAS can support your roadmap.

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*This demo is part of our Reference Architecture Series – built to inspire smarter test development.*



# How to Run the Adaptive Test Demo

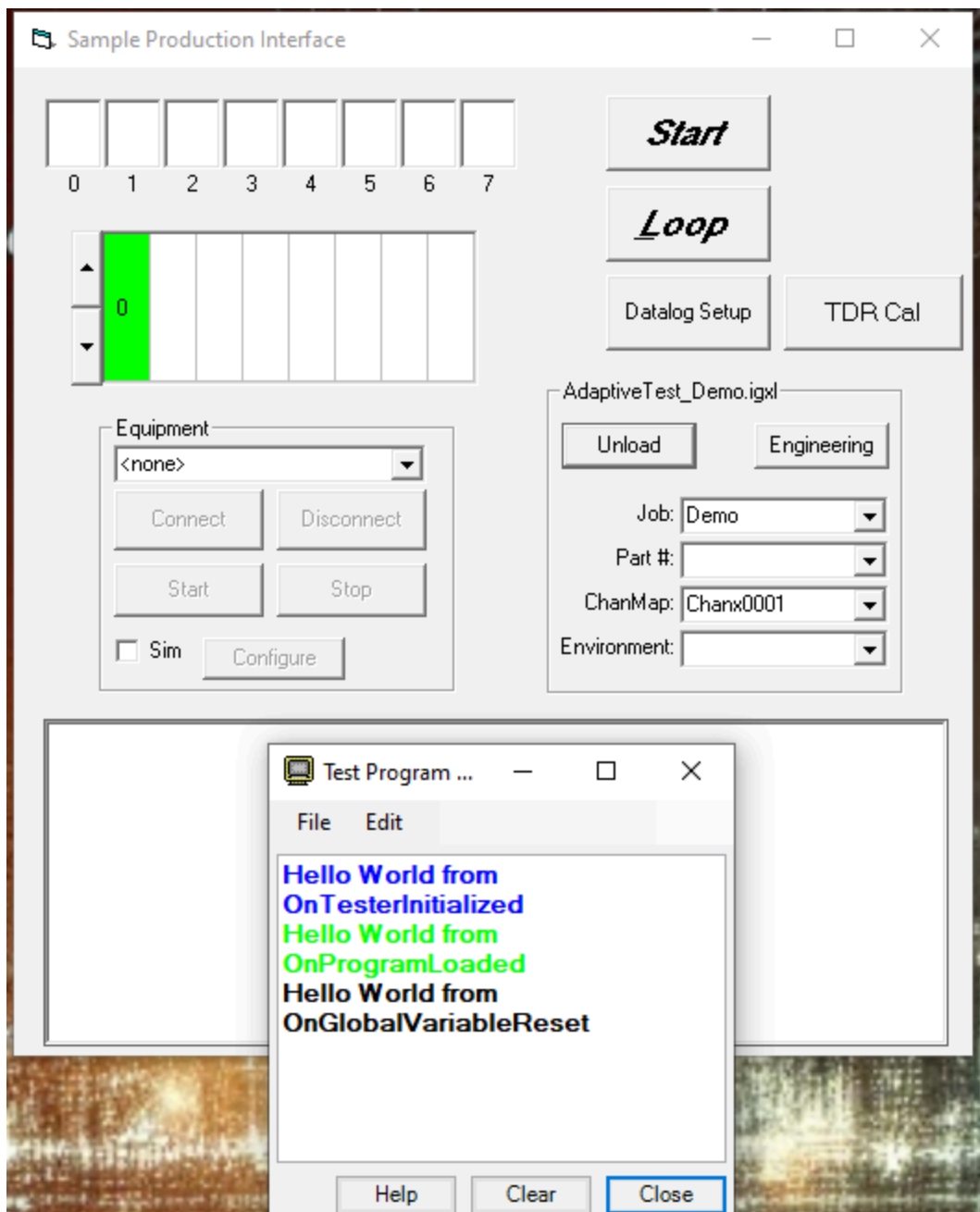
Reference: RA\_453

This section provides step-by-step instructions to explore the key features of the Adaptive Test DAS Demo.

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## Step 1: Load the Test Program

Use the SimpleOI interface to load your test program.



## Step 2: Execute the Program in Loop Mode

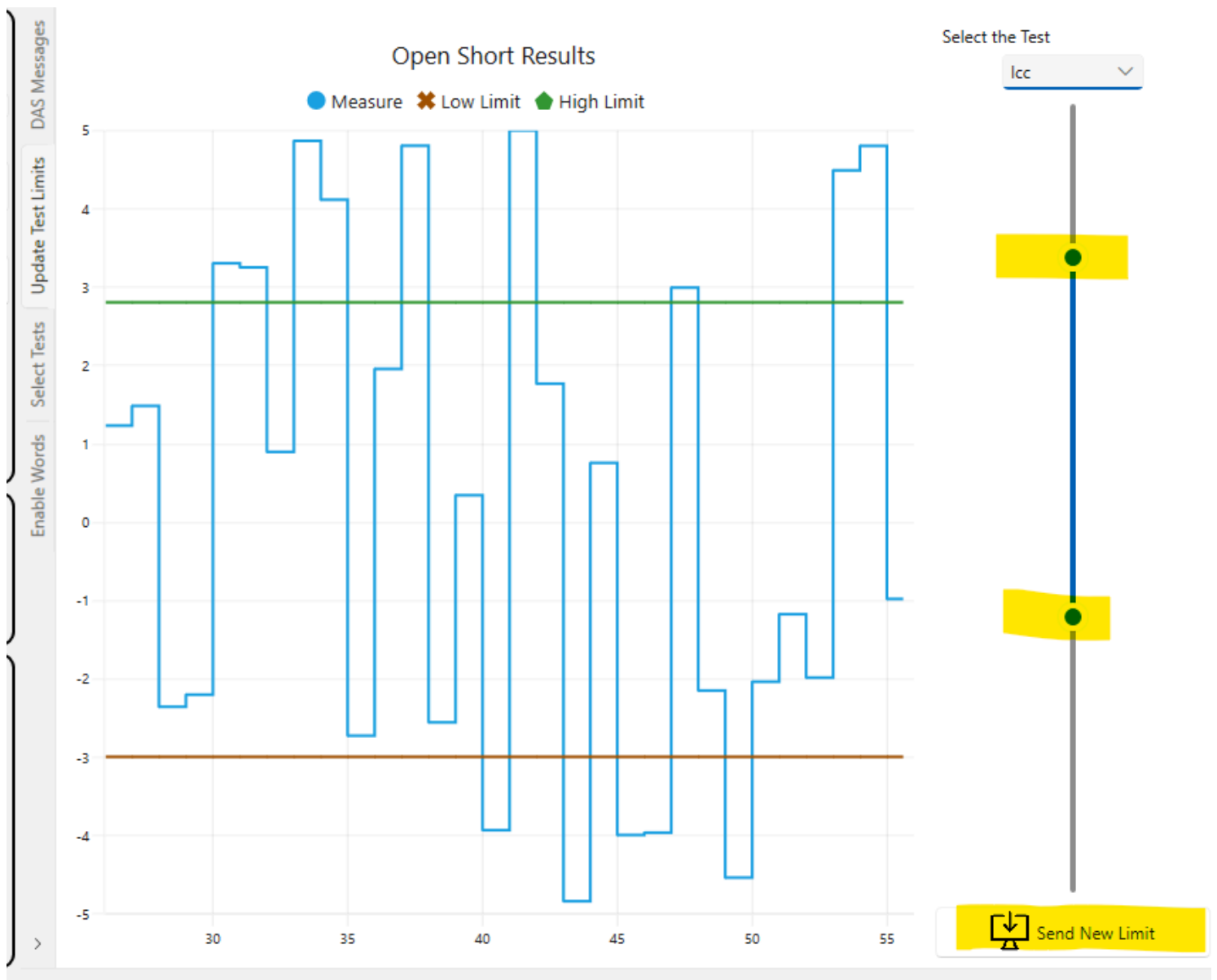
Click the **Loop** button to execute the program 10 times. Once completed, press the **Stop** button.



## Step 3: Adjust Test Limits Dynamically

In the demo application, click **Update Test Limits**. Use the slider to modify the thresholds, then click **Send Limits** to apply them.





Run the test a few times from the SimpleOI interface to observe the impact.

## Step 4: Observe the Updated Limits

The graphical output will reflect the new limit values.

**DIA**

Adaptive Tests

**TERADYNE**

## Communication Status

DAS URL

<http://127.0.0.1:3000/tems/>

Connect the DAS



Disconnect the DAS



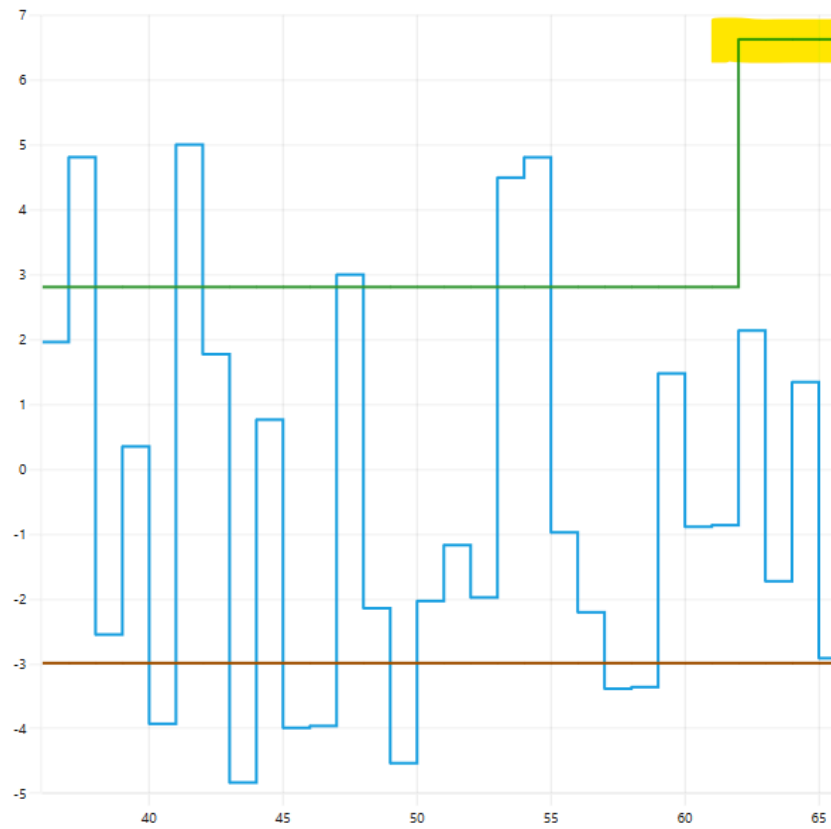
Clear Adaptive Command

Adaptive Command

DAS Messages  
Update Test Limits  
Select Tests  
Enable Words

## Open Short Results

● Measure ✕ Low Limit ◆ High Limit



Select the Test

lcc



Send New Limit

In DAS, select the **Adaptive\_Change limits** message. Click on the message to inspect its contents.

| Time Stamp    | Test Cell | Name            | Content                                                                                                          |
|---------------|-----------|-----------------|------------------------------------------------------------------------------------------------------------------|
| 08:01:48.9137 | TEMS-01   | ADAPTIVE_CHANGE | TEP_ACTIONS":<br>[{"ACTION":"UPDATE_LIMITS","REQUEST_ID":1,"TESTSTEP_NAMES":<br>["Icc"],"LIMITS":{"LIMIT_NAMES": |

```
{
  "TIME_STAMP": "2025-05-05T08:01:48.9137009Z",
  "LOT_ID": "UNKNOWNLOT_PID_8380_1",
  "SUBLOT_ID": "UNKNOWNSUBLOT_PID_8380_1",
  "DAS_ID": 1,
  "TESTSTEP_ACTIONS": [
    {
      "ACTION": "UPDATE_LIMITS",
      "REQUEST_ID": 1,
      "TESTSTEP_NAMES": [
        "Icc"
      ],
      "LIMITS": {
        "LIMIT_NAMES": [
          "Icc"
        ]
      }
    }
  ]
}
```

You can also observe the received message in the IGXL test program output.

Teradyne\_IG-XL\_DataCollect\_8380 - OutputWin
File Edit Collect View Help

Site
Sort
Bin

0
1
1

OnProgramEnded: 2025-05-05 10:01:40.390
Adaptive Change: The limits {Icc} on test step {Icc} were changed. New low limit = -3. New high limit = 6.61954453733064.
OnProgramStarted: 2025-05-05 10:01:48.910
Device#: 63

| Number     | Site | Test Name | Pin | Channel | Low           | Measured     | High         | Force  | Loc |
|------------|------|-----------|-----|---------|---------------|--------------|--------------|--------|-----|
| <Icc_Fake> |      |           |     |         |               |              |              |        |     |
| 1000       | 0    | Icc       | -1  |         | -3000.0000 mA | 2127.3046 mA | 6619.5445 mA | 0.0000 | 0   |

## Step 5: Work with Enable Words

This part demonstrates how to activate a sub-flow using enable words. Initially, only the test **ICC** is active.

To explore more:

- Access the engineering view in SimpleOI (password: **Teradyne**)
- Open the test program and locate the **DemoFlow**
- Visualize the flow and locate the enable word logic

[illegible]

Click on the **OnePin EnableWord** option and run the test program once.

Select the enable words you want to enable or disable

☒ Enable or Disable the enable word : OnePin

☐ Enable or Disable the enable word : DualPin

You'll see that more tests are now activated:

Adaptive Change: The following enable words were enabled: {OnePin}.

OnProgramStarted: 2025-05-05 10:09:38.800  
Device#: 68

| Number                | Site | Test Name       | Pin | Channel | Low                 | Measured            | High          | Force  | Loc |
|-----------------------|------|-----------------|-----|---------|---------------------|---------------------|---------------|--------|-----|
| <Icc_Fake>            |      |                 |     |         |                     |                     |               |        |     |
| 1000                  | 0    | Icc             | -1  |         | -3000.0000 mA       | 833.5900 mA         | 6619.5445 mA  | 0.0000 | 0   |
| OnePin                |      |                 |     |         |                     |                     |               |        |     |
| <OnePin_NoHighLimit>  |      |                 |     |         |                     |                     |               |        |     |
| 200                   | 0    | OnePin_NoHighLi | -1  |         | 100.0000 m          | 1.1235              | N/A           | 0.0000 | 0   |
| <OnePin_NoLowLimit>   |      |                 |     |         |                     |                     |               |        |     |
| 300                   | 0    | OnePin_NoLowLim | -1  |         | N/A                 | 500.0000 m          | 900.0000 m    | 0.0000 | 0   |
| <OnePin_NoLimit>      |      |                 |     |         |                     |                     |               |        |     |
| 400                   | 0    | OnePin_NoLimit  | -1  |         | N/A                 | -700.0000 m         | N/A           | 0.0000 | 0   |
| <OnePin_MilliAmpUnit> |      |                 |     |         |                     |                     |               |        |     |
| 500                   | 0    | OnePin_MilliAmp | -1  |         | -1000000000.0000 nA | 17500000000.0000 nA | (F) 0.0000 nA | 0.0000 | 0   |

A new Adaptive Change message will also be generated:

|               |         |                 |                                                                                                                                                           |
|---------------|---------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08:09:38.8061 | TEMS-01 | ADAPTIVE_CHANGE | ID_8380_1","SUBLOT_ID":"UNKNOWNSUBLOT_PID_8380_1","DAS_ID":1,"TESTSTEP_ACTIONS":[],"ENABLEWORD_ACTIONS":[{"ACTION":"ENABLE","REQUEST_ID":2,"ENABLEWORDS": |
|---------------|---------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|

```

"TIME_STAMP": "2025-05-05T08:09:38.806154Z",
"LOT_ID": "UNKNOWNLOT_PID_8380_1",
"SUBLOT_ID": "UNKNOWNSUBLOT_PID_8380_1",
"DAS_ID": 1,
"TESTSTEP_ACTIONS": [],
"ENABLEWORD_ACTIONS": [
  {
    "ACTION": "ENABLE",
    "REQUEST_ID": 2,
    "ENABLEWORDS": [
      "OnePin"
    ],
    "STATUS": "OK"
  }
]

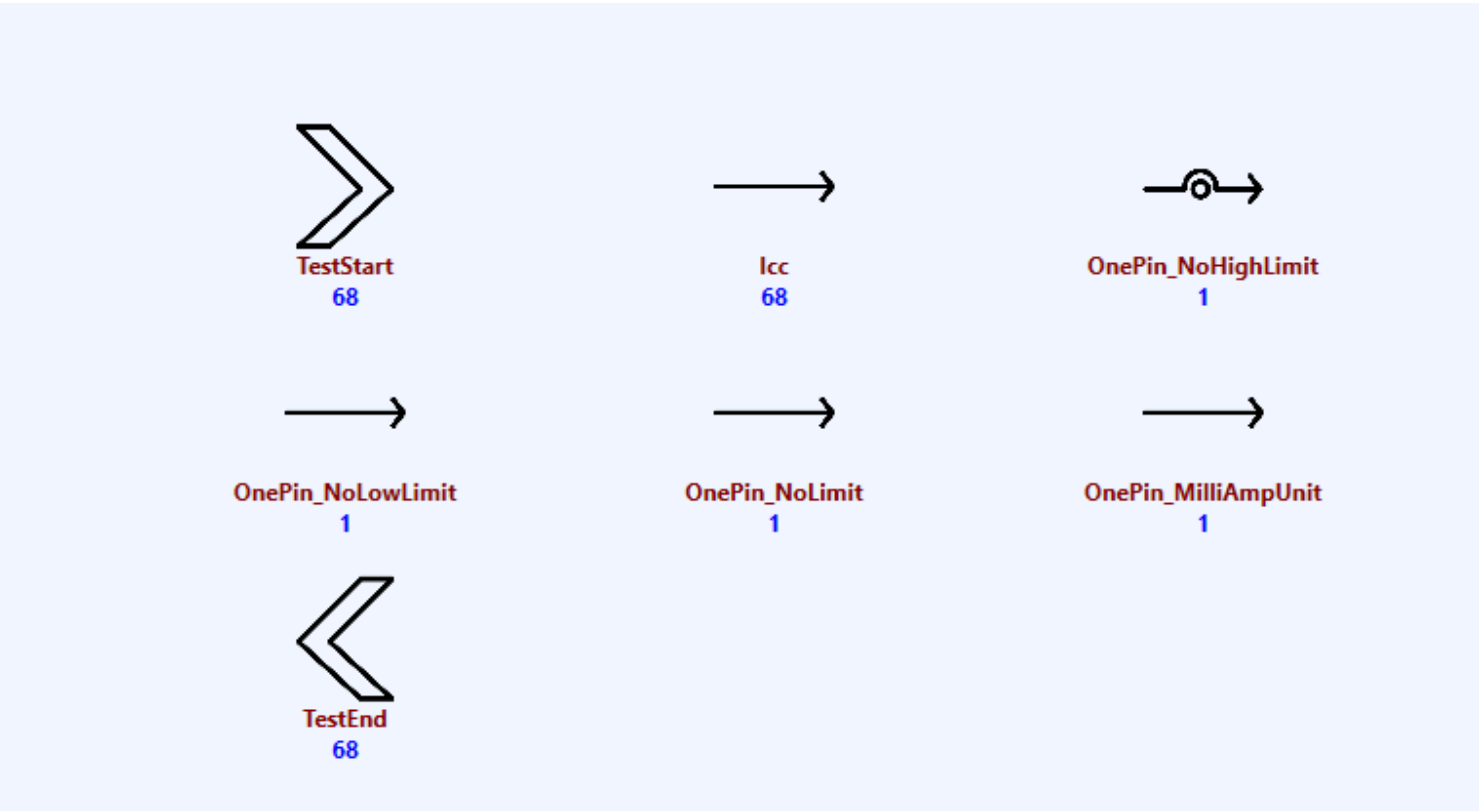
```

## Step 6: Enable or Disable Tests

This step shows how individual tests can be selectively enabled or disabled.

- Go to the **Select Test** tab in the DAS interface
- A count of executions for each test is shown in blue
- For best demo results, activate **DoAll** in IGXL Run Options

Now, disable the `OnePin_NoHighLimit` test by selecting it.



Run the test program once. You'll see that the test is skipped.

|                                                                                                                                                                                                                                                                                                                                         |         |                 |                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 08:25:20.0344                                                                                                                                                                                                                                                                                                                           | TEMS-01 | ADAPTIVE_CHANGE | ID_8380_1";SUBLOT_ID":"UNKNOWNSUBLOT_PID_8380_1";"DAS_ID":1;"TEST<br>TEP_ACTIONS":[{"ACTION":"DISABLE","REQUEST_ID":3,"TESTSTEP_NAMES": |
| <pre>{   "TIME_STAMP": "2025-05-05T08:25:20.034446Z",   "LOT_ID": "UNKNOWNLOT_PID_8380_1",   "SUBLOT_ID": "UNKNOWNSUBLOT_PID_8380_1",   "DAS_ID": 1,   "TESTSTEP_ACTIONS": [     {       "ACTION": "DISABLE",       "REQUEST_ID": 3,       "TESTSTEP_NAMES": [         "OnePin_NoHighLimit"       ]     }   ],   "STATUS": "OK" }</pre> |         |                 |                                                                                                                                         |

You can confirm this in IGXL:

OnProgramEnded: 2025-05-05 10:09:38.870  
Adaptive Change: The following test steps were disabled: {OnePin\_NoHighLimit}.  
OnProgramStarted: 2025-05-05 10:25:20.030  
Device#: 69

| Number                | Site | Test Name       | Pin | Channel | Low                 | Measured                | High         | Force  | Loc |
|-----------------------|------|-----------------|-----|---------|---------------------|-------------------------|--------------|--------|-----|
| <Icc_Fake>            |      |                 |     |         |                     |                         |              |        |     |
| 1000                  | 0    | Icc             | -1  |         | -3000.0000 mA       | -4192.8535 mA (F)       | 6619.5445 mA | 0.0000 | 0   |
| OnePin                |      |                 |     |         |                     |                         |              |        |     |
| <OnePin_NoLowLimit>   |      |                 |     |         |                     |                         |              |        |     |
| 300                   | 0    | OnePin_NoLowLim | -1  |         | N/A                 | 500.0000 m              | 900.0000 m   | 0.0000 | 0   |
| <OnePin_NoLimit>      |      |                 |     |         |                     |                         |              |        |     |
| 400                   | 0    | OnePin_NoLimit  | -1  |         | N/A                 | -700.0000 m             | N/A          | 0.0000 | 0   |
| <OnePin_MilliAmpUnit> |      |                 |     |         |                     |                         |              |        |     |
| 500                   | 0    | OnePin_MilliAmp | -1  |         | -1000000000.0000 nA | 17500000000.0000 nA (F) | 0.0000 nA    | 0.0000 | 0   |

# Download the Demo

[Download the demo from here](#)